Lab 3: Minesweeper — Hexagonal Mode & Interaction Refinements

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Purpose and relationship to Lab 2

This document describes the work completed in Lab 3, building on the playable rectangular Minesweeper game from Lab 2. The original Lab 2 report is included separately and documents the Simple/Advanced modes, core reveal and marking mechanics, recursive expansion, and initial undo implementation. This report focuses on the new hexagonal mode and subsequent interaction improvements.

Innovative mode: a playable hexagonal board

- Third difficulty: Innovative mode uses a 10 × 10 board with 20 mines, alongside the existing Simple (9 × 9, 15 mines) and Advanced (12 × 12, 30 mines) modes.
- Custom rendering: HexBoardUi draws pointy-top hexagons using Graphics2D and polygon vertices computed with trigonometry. Alternating rows are horizontally offset.
- Mouse interaction: polygon hit testing (Polygon.contains) identifies the hexagon under the pointer, enabling left-click reveal and right-click flag/unflag.
- Hexagonal game rules: BoardModel calculates row-parity-dependent neighbors for the offset grid. Each hexagon has up to six neighbors, compared with up to eight on the rectangular board.
- Shared game behavior: safe first reveal, recursive empty-area expansion, mine counts, flags, timer, reset, and win/loss detection also work in Innovative mode.

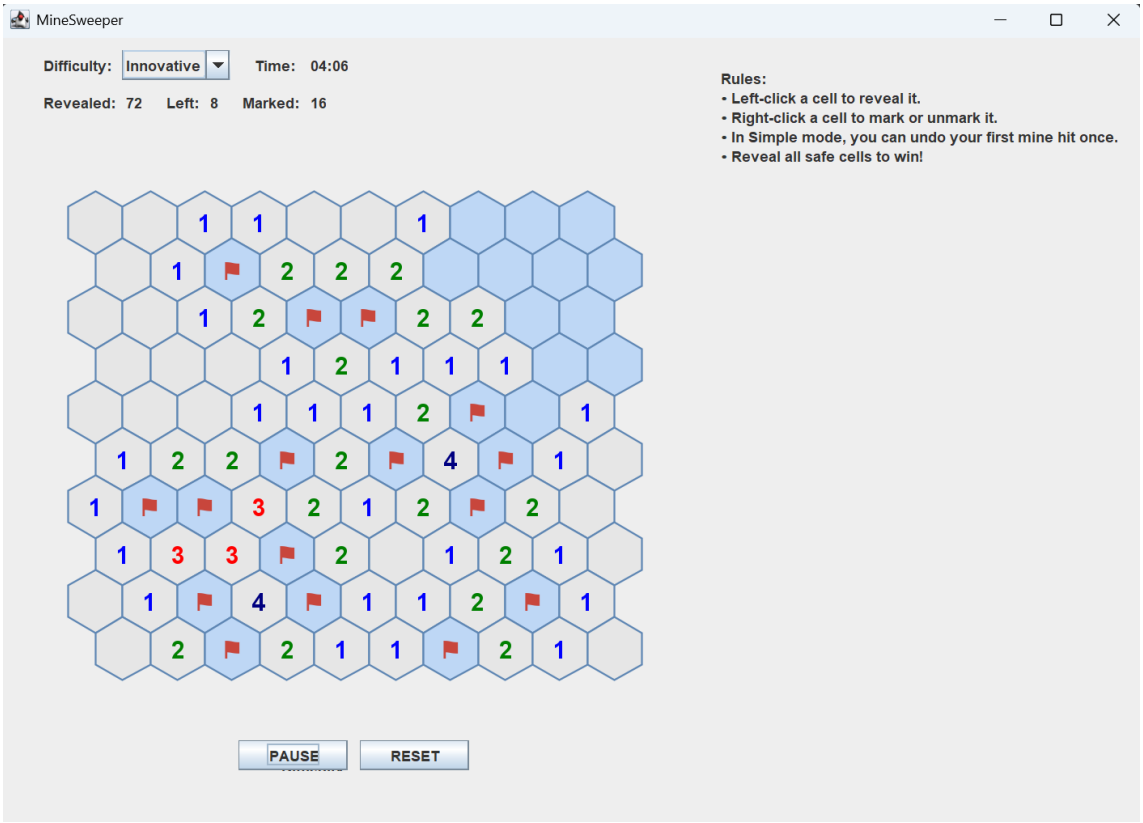


Figure 1. Innovative mode: hexagonal geometry, numbered cells, and flags during play.

Interaction improvements

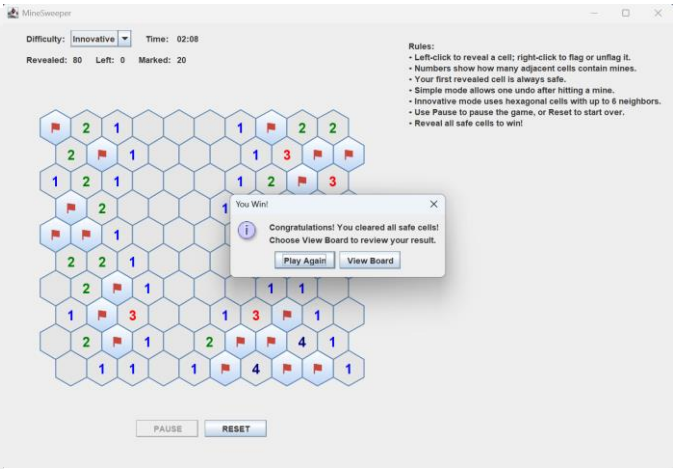
- Pause / Resume: a translucent “GAME PAUSED” overlay covers the board, intercepts input, and pauses the elapsed-time counter.
- End-of-game review: the updated dialog offers Play Again and View Board; the latter preserves the finished board for inspection.
- Clearer cell states: revealed rectangular cells, displayed mines, and the finished board no longer show misleading button-press feedback; number colors remain intact.
- Undo compatibility: Simple mode still allows one undo after the first mine hit, with the affected cell restored to normal interaction afterward.

Interaction States & Validation of Lab 3

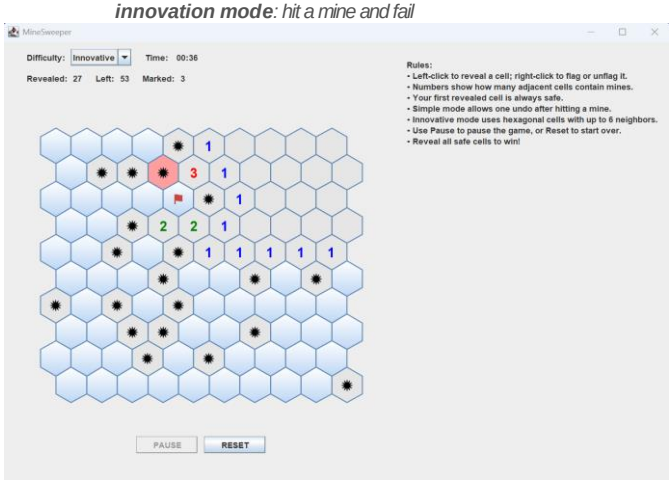
Screenshots from the final implementation



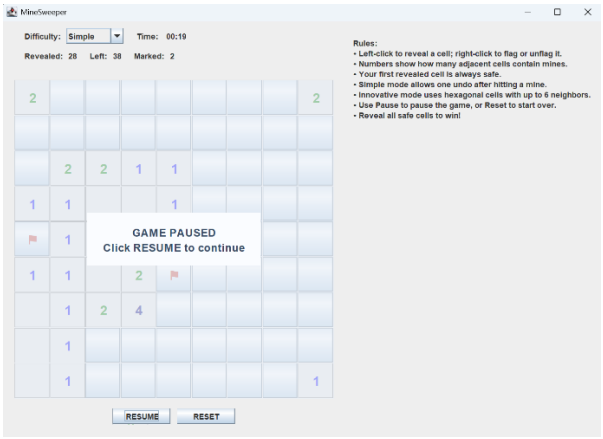
***innovation mode**: hit a mine and fail*



***innovation mode:** win the game*



***innovation mode:** view the board after the game*



***simple mode:** pause overlay and RESUME control.*

Testing and validation

I manually tested all three modes, switching difficulty, safe first clicks, rectangular and hexagonal neighbor counts, automatic reveal, flag/unflag, counters, pause/resume, reset, one-time Simple undo, mine-hit loss, victory, and game-over board review. I also checked that revealed cells do not show a pressed state and that Undo restores normal clicking.

Packaging

The final Java Swing application is organized into model, controller, and ui packages and has been packaged as an executable JAR, successfully launched independently of IntelliJ using Java 21. The Lab 2 PDF is provided separately to document the original implementation.

Conclusion

In Lab 3, I extended the Minesweeper game developed in Lab 2 by implementing an innovative hexagonal board and improving the overall user interaction. The new mode required custom rendering, mouse hit detection, and a six-neighbor calculation algorithm. Additional refinements improved the clarity and consistency of game feedback across different states. All three game modes were tested, and the final application was successfully packaged as an executable JAR.